INCH-POUND

MIL-P-53087(ME)
3 February 1989

MILITARY SPECIFICATION

POWER PLANT, ELECTRIC,

DIESEL ENGINE DRIVE, TRAILER MOUNTED,

3 KW, 60 HERTZ, AN/MJQ-32

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the trailer-mounted generator set (hereinafter referred to as a power plant) designated AN/MJQ-32. The power plant consists of two Department of Defense standard, diesel-engine-driven, 3 KW, 60 hertz generator sets, two acoustic suppression kits (ASK), various auxiliary equipment, brackets, and hardware mounted on a modified Government-furnished trailer chassis.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6115

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-P-291	- Paperboard, Wrapping and Cushioning.
PPP-T-76	- Tape, Pressure Sensitive Adhesive, Packaging of.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-E-10062	- Engine, Preparation for Shipment and Storage of.
MIL-B-22191	- Barrier Material, Transparent, Flexible, Heat Sealable.
MIL-G-28554	- Generator Sets, Mobile Electric Power and Supplemental Equipment; Packaging of.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-R-46164	- Rustproofing for Military Vehicles and Trailers.
MIL-G-53082	- Generator Sets, Diesel Engine Driven 3 KW, 60 Hertz, 400 Hertz, and 28 Volt Direct Current, Type I (Tactical), Class 2 (Utility).

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-705	- Generator Sets, Engine-Driven, Methods of Test and Instructions.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing.
MIL-STD-1474	- Noise Limits for Army Materiel.
MIL-STD-1791	- Design for Internal Aerial Delivery in Fixed Wing Aircraft.
MS52152	- Chassis, Trailer: 3/4 Ton, 2 Wheel, Military Design, M116A2.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13228E9895 - Power Plant, Electric, AN/MJQ-32.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATIN: SIRBE- , Fort Belvoir, VA 22060-5606.)

THE MILITARY TRAFFIC MANAGEMENT COMMAND
Transportation Engineering Agency (MTMC/TEA)

Pamphlet 70-1 - Transportability for Better Strategic Mobility.

(Application for copies should be addressed to The Military Traffic Management Command, Transportation Engineering Agency (MTMC/TEA), ATIN: MTT-TR, P. O. Box 6276, Newport News, VA 23606-0276.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN WELDING SOCIETY (AWS)

B2.1 - Welding Procedure and Performance Qualification

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJune Road, P.O. Box 351040, Miami, FL 33135.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open-Top Cars.

(Application for copies should be addressed to the Association of American Railroads, American Railroads Building, 50 F Street, NW, Washington, DC 20001.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The power plants, AN/MJQ-32, shall be in accordance with Top Assembly TA13228E9895 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The power plant shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the power plant may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.1.3 Rust proofing. The trailers shall be rust proofed in accordance with MIL-R-46164.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the power plant may be newly fabricated from recovered materials to the maximum extent practicable,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

provided the power plants produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the power plants.

3.4 Noise. The noise levels shall not exceed MIL-STD-1474, 5.1.1.1, category D [85 dB(A)], at the operator's position or any other occasionally occupied positions, when tested in accordance with 4.5.2.4. If noise levels exceed 85 dB(A), then the precautions of MIL-STD-1474 shall be provided. Hazard signs shall conform with MIL-STD-1474, 4.3, and be readable at the distance indicated on the sign.

3.5 Safety. Any changes, deviations, and waivers to TA13228E9895 shall conform with the following requriements. All rotating or reciprocating parts and other parts subject to high operational temperatures that are of such a nature or are so located as to be a hazard to the operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. All surfaces that can be used for walking or standing shall have a non-slip surface attached.

3.6 Vehicle marking. Registration numbers and other marking of the trailer shall be in accordance with MIL-STD-642 (see 6.10).

3.7 Government-furnished property. The following property in the quantities indicated will be furnished by the Government (see 6.4):

Item No.	Description	Identification	Qty for each AN/MJQ-33
1.	Chassis, Trailer: 3/4 Ton, 2 Wheel, Model M116A2	MS52152	1
2.	Generator Set, Diesel-engine-driven, 3 KW, 60 Hertz	MIL-G-53082	2

3.8 Workmanship. Workmanship shall be of a quality to assure that the power plants are free from defects resulting from defective material, incorrect manufacturing or assembly practices, incorrect treatment and painting, incomplete welds, rust, cracks, and other defects that could impair the plants' operation or serviceability. External surfaces shall be free from burrs, sharp edges and corners except when sharp edges and corners are specified on the applicable drawing. Welding procedures and welder performance qualification shall be in conformance with AWS B2.1.

3.9 Transportability. The power plant shall be capable of being transported by military or commercial trailers, trains, vessels, aircrafts and can withstand the impact forces encountered in shipment without damage or permanent deformation. The power plant shall be equipped with tiedown and slinging provisions.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:12Z
Check the source to verify that this is the currentversion before use.

Marking shall be in accordance with MIL-STD-129. 3.4 Marking.

3.9.1 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type II equipment, and to MIL-STD-1791 for equipment resistant criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

3.9.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, type II. The provisions shall enable the complete power plant to be lifted, in the normal operating position. The provisions shall be located so that not less than one inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. SLinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.9.1. All slinging/tiedown provisions shall be labeled 'LIFT', 'TIEDOWN', or 'LIFT TIEDOWN', as appropriate, in 1-inch (2.54 cm) high letters.

3.9.3 Air transportability. The contractor shall provide sufficient data to the Government in the technical information package to confirm that the power plant being supplied can be loaded and transported in the C-130, C-141, and C-5A aircraft. The power plant shall meet the requirements of MIL-STD-1791 for air transport. Disassembly to achieve air transportable configuration shall be accomplished in not more than 2 hours, and reassembly shall be accomplished in not more than 2 hours, using only 2 people and common tools, and lifting devices available in the military unit.

3.9.4 Rail transportability. The power plant shall be rail transportable in CONUS and NATO countries without restrictions. The power plant shall have a dimensional profile within the Gabarit International de Chargement (GIC) in accordance with MTMC/TEA pamphlet 70-1, outline diagram when loaded on a 50-inch (127 cm) high rail car. The power plant shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.8.3 without failure, damage, or permanent deformation.

3.9.5 Highway transportability. The power plant, when loaded on a semitrailer/tractor, shall be within the highway permit limits of all States.

3.10 Switch box. Provisiong shall be made for switching load from one generator set to the other, while both sets are running. To accommodate the generator set 5 wire systems the switch box shall have a 5th terminal that is connected via a low impedance connection to the box. This 5th terminal, in addition to the (4) load terminals, shall be of sufficient size (maybe multiple terminals) to accept the 5th wire from each generator set and from the load and to accept a ground rod cable.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Disassembly inspection. Failure of any examination or test of the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.4 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings.

4.1.5 Acceptability criteria. Power plants which conform to all requirements in section 3 and 5 and pass all applicable examinations and tests in section 4 of this specification will be considered acceptable by the Government.

4.2 Classification of inspections. The inspections requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

The contractor should apply for nomenclature in accordance with the applicable clause

4.3 First article inspection.

4.3.1 Examination. The first article power plant shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. Power plant components shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Examination. Each power plant shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Power plant components shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The power plants shall be examined for the following defects:

101. Materials not as specified.
102. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the power plants.
106. Safety not as specified.
107. Vehicle registration marking and identification plate missing, incomplete, improper location or size.
108. Treatment and painting not as specified.
109. Dimensions not as shown on drawings.
110. Components missing or not as specified.
111. Assembly not as shown on drawings.
112. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. The following preparations shall be made prior to testing the switchbox assembly:

a. Cables running between the switchbox assembly and the generator sets shall be installed to the switchbox assembly rotary switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

b. The free ends of the power input cables shall be attached to the generator sets as follows:

(1) Cable wires marked L1 shall be attached to Phase A.
(2) Cable wires marked L2 shall be attached to Phase B.
(3) Cable wires marked L3 shall be attached to Phase C.
(4) Cable wires marked L0 shall be attached to neutral.

c. Provide ground to ground terminal of switchbox.
d. A 3-phase, 4-wire, 3 KW resistive load shall be connected to the output of the switchbox.

4.5.2.2 Test procedures. Tests shall be conducted in accordance with MIL-STD-705 and as specified herein. Test instruments shall be of the laboratory type and shall have been calibrated within 30 days prior to the start of testing and at intervals not greater than 6 months thereafter.

4.5.2.3 Operation. The rotary switch of the switchbox shall be placed in the generator No. 1 position. Start and operate generators in accordance with operator's manual at the rated load through the rotary switch for not less than 5 minutes. During this period, phase sequence (rotation) shall be tested in accordance with MIL-STD-705, method 507.1. Phase rotation shall be L1, L2, and L3. This test shall be performed at the load terminals of the switchbox. Phase rotation not as specified shall constitute failure of this test. Failure of the GFE generators to operate in accordance with the operator's manual shall not constitute failure of this test. The contracting officer shall be notified immediately of any such failure.

4.5.2.4 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack.

4.6.1.1 Examination. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of packaging.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Sample power plants selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

113. Preservation not as specified for all levels (see 5.2).
114. The bottom of the generators set not protected with plywood shields as specified for level A (see 5.3.1).
115. Housing for the power plant not as specified for level A or level B (see 5.3.1 and 5.3.2).
116. Strapping not as specified for level A or level B (see 5.3.1 and 5.3.2).
117. Accessory and lose parts not packed as specified for all levels (see 5.3).
118. Blocking, bracing and anchoring not as specified for levels A and B (see 5.3).
119. Landing gear(s) on the trailer(s) are not up as specified for all levels (see 5.3).
120. Marking incorrect, illegible or missing (see 5.4).
121. Depreservation guide not preserved and secured as specified (see 5.5).

4.7 Inspection of transportability.

4.7.1 Quality conformance inspection of transportability.

4.7.1.1 Examination. Each power plant shall be examined for the following defects. AQL shall be 1.0 percent defective.

123. Tiedown provisions not as specified (see 3.9.1).
124. Slinging provisions not as specified (see 3.9.2).
125. Air transportability requirements not met (see 3.9.3).
126. Rail transportability requirements not met (see 3.9.4).
127. Highway transportability requirements not met (see 3.9.5).

4.7.2 Transportability tests.

4.7.2.1 Tiedown provisions. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.9.1. Inability to meet the requirements of 3.9.1 shall constitute failure of this demonstration.

4.7.2.2 Slinging provisions. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.9.2. Inability to meet the requirements of 3.9.2 shall constitute failure of this demonstration.

4.7.2.3 Rail impact test. The first article power plant shall be tested in accordance with the following rail impact test procedure:

Step 1. From one to five railcars shall be positioned on a level section of track. The following conditions shall be satisfied:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

a. The total weight of the car(s) shall be at least 250,000 pounds (114,000 kilograms).
b. The couplers between any cars shall be compressed to take up any slack.
c. All the air and hand brakes on the car(s) used shall be set.
d. Any load in or on the car(s) shall be secured to prevent sliding or shifting; any movement greater than 1.97 inches (5 centimeters) shall be justification for retest.
e. The end of the buffer car to be struck must have a standard draft gear.

Step 2. Mount the test item on the test car. The test car shall be equipped with standard draft gears and conventional under frames. Mounting of the test item shall incorporate the standard loading and bracing method as shown in section 6 of the Association of American RailRoads (AAR) "Rules Governing the Loading of Department of Defense Materiel on the Open Top Cars." No exotic or unusual tiedown methods shall be used; any non- standard loading and bracing must be approved by the MTMC/TEA prior to testing. The arrangement of the test item and its blocking and tiedown to be tested shall be identical to that proposed and approved by MTMC/TEA (if nonstandard).

Step 3. Situate the test car between the buffer cars and the locomotive, and pull the test car at least 200 feet (65 meters) from the buffer car(s) along a level section of track (a minimum distance to achieve the required locomotive speeds).

Step 4. Position the knuckles of the buffer and test cars for coupling.

Step 5. Install a timing device to measure the test car speed (+ or = .1 km/hr) just prior to impact with the buffer car(s). Suggested methods include electronic timing (microswitches) and radar. The use of torpedoes or a stopwatch is permissible but not recommended because of the inaccuracies involved.

Step 6. Push the test car toward the buffer car(s) and, by using the locomotive's speedometer or other means, release the test car when the desired test speed is reached, thus allowing the test car to freely impact the buffer car(s).

Step 7. Repeat step 6 until the test car(s) has impacted the buffer car three times at the same end, once each at speed of 4, 6, and 8 mph (6.4, 9.7 and 13 km/hr), + 0.5 mph (0.8 km/hr) or - 0.0 mph (0.0 km/hr). Reverse the test car and repeat the 8 mph (13.0 km/hr) for a total of 4 impacts.

Note: Adjustment of the lading or securing mechanism or reconditioning of the bracing or items of securement are not allowed without a complete retest.

4.7.2.4 Failure criteria. Failure to meet the requirements of 3.9.4 and 4.7.2.3 rail impact test shall constitute failure of this test.

5. PACKAGING

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the government (see 6.3.1). The first article pack may be prepared utilizing either the first article model or a production model. When the first article model is utilized, any preservation and packing shall be removed by the contractor at no expense to the government, when requested by the government, to facilitate comparison between the first article model and production model.

5.2 Preservation. Preservation shall be level A, B or level C (see 6.2).

5.2.1 Level A. The power plant, and accessory equipment, shall be preserved in accordance with the level A preservation requirements of MIL-G-28554. Each generator set shall be further preserved in accordance with the requirements for method IIa of MIL-P-116 as specified in MIL-G-28554 for sheathed crated sets. The generator set shall be secured directly through the barrier material to the chassis using the required mounting bolts. Any openings in the trailer bed, through which an objective may enter from below and cause damage to the method IIa barrier material, shall be completely closed off the application of a plywood shield(s). Such shield(s) shall be constructed of 0.50-inch (1.27 cm) thick plywood meeting the requirements of NN-P-530, group B, interior type (exposure 1) with the exterior (waterproof) glue, grade CD, and shall be securely attached to the bottom of the trailer bed. Internal type humidity indicators are required with inspection windows located in the barrier material for ease of access to the indicator. Cooling systems for engines with liquid coolant systems shall be preserved in accordance with the preservative and drain procedure of MIL-E-10062 and tagged accordingly. Before sealing the barrier material, the sides, ends and top of each generator set shall be wrapped with a double thick wrap of cushioning material conforming to PPP-P-291, type I, style 2, to protect the barrier material from any sharp edges or protrusions.

5.2.2 Level B. The power plant, and accessory equipment, shall be preserved in accordance with level B requirements of MIL-G-29554. Cooling systems for engines with liquid coolant systems shall be filled with equal parts of clean water, and antifreeze conforming to MIL-A-46153.

5.2.3 Level C. The power plant and accessory equipment shall be preserved in accordance with level C of MIL-C-28554. Cooling systems for engines with liquid coolant systems shall be filled with equal portions of clean water and antifreeze conforming to MIL-A-46153.

5.3 Packing. Packing shall be level A, B, or C (see 6.2).

5.3.1 Level A. Each power plant and accessory equipment, preserved as specified in 5.2, shall be packed in a box type housing that consisting of ends, sides and top, constructed in accordance with PPP-B-601, overseas type, style I. Sides and ends shall be 0.25 inches (0.635 cm) thick plywood and the top shall be 0.50 inches (1.27 cm) thick plywood. The housing shall be fasteners, wood wedges

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

and metal strapping. Such means of fastening shall not damage the method IIa barrier material. Metal strapping shall conform to QQ-S-781, type I or IV, finish B. The PPP-B-601 housing may be of the size necessary to encompass both generator sets as a single housing or it may consist of two separate housing each of the size necessary to encompass an individual generator set. The preserved accessory equipment shall be positioned, blocked, anchored and braced within the single housing in accordance with MIL-STD-1186. When two separate housings are utilized, preserved accessory equipment for which space is not available within the box type housing or generator set housing shall be provided additional protection by placing them in snug-fitting boxes conforming to PPP-B-601, overseas type. Cushioning, blocking and bracing shall be applied to the contents, as applicable, to prevent movement within the boxes. The boxes shall be secured to the trailer.

Caution: Landing gear(s) on the trailer(s) are subjected to damage during shipment. Do not ship trailer(s) with landing gear(s) down.

5.3.2 Level B. The power plant and accessory equipment, preserved as specified in 5.2, shall be packed for level B in the same manner as specified for level A in 5.3.1 except for the following:

a. The PPP-B-601 housing shall be domestic type.
b. The QQ-S-781 metal strapping shall be finished A.

Caution: Landing gear(s) on the trailer(s) are subjected to damage during shipment. Do not ship trailer(s) with landing gear(s) down.

5.3.3 Level C. The power plant, preserved as specified in 5.2, shall be packed in accordance with MIL-G-28554, level C.

Caution: Landing gear(s) on the trailer(s) are subjected to damage during shipment. Do not ship trailer(s) with landing gear(s) down.

5.4 Marking. In addition to any special or identification marking required by the contract of purchase order (see 6.2), each power plant shall be marked in accordance with MIL-STD-129.

5.5 Depreservation guide. The depreservation guide shall be preserved in accordance with MIL-P-116, method IC-1; however, barrier material shall be in accordance with MIL-B-22191, type I, and heat sealed. It shall be secured on the generator set with tape in accordance with PPP-T-76.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The power plant is intended for use by the Armed Services to supply electrical power for military operations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval and number of units required (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. Time frame required for submission of the first article pack (see 5.1).
f. When transportability report is required (see 6.7).
g. Level of preservation and packing required (see 5.2 and 5.3).
h. Any special markings (see 5.4).
i. When other than DA Form 2258 is used (see 6.8).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.7.

6.5 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

" The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:12Z
Check the source to verify that this is the current version before use.

6.6 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the power units in accordance with this specification.

6.7 Transportability report. When specified (see 6.2), the contractor will furnish a transportability report in accordance with AR 70-47 and S.O.P 70-38 for the power plant, 95 days prior to delivery of the first article power plant.

6.8 Depreservation guide. A depreservation guide should be prepared and provide, unless otherwise specified (see 6.2), DA Form 2258 "Depreservation Guide for Vehicles and Equipment" should be used (see 5.5).

6.9 Definition.

6.9.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP's indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.10 Registration number. The contracting officer should arrange to furnish a list of registration numbers to be applied to the trailer (see 3.6).

6.11 Subject term (key word) listing.

Cable wires
Generator sets
Power plant
Registration numbers
Trailer chassis

Custodian:

Army - ME

Preparing activity:

Army - ME

Project 6115-A532

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-53087(ME)	Power Plant, Electric, Diesel Engine Drive, Trailer Mounted, 3 KW, 60 Hertz, AN/MJQ-32

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:12Z
Check the source to verify that this is the currentversion before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
USA Belvoir RDE Center
STRBE-TSE
Fort Belvoir, VA 22060-5606

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.